Digital Electronics

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Digital Electronics (A11147)

Short Title: Digital Electronics
Faculty Science and Computing
Department Computing and Mathematics
Cluster Automotive, Automation and IoT
Author EREADE
Credits 5 **Level:** Introductory

Description of Module / Aims

The logical outputs of digital circuits are determined by the logical function being performed and the logical input states at that particular moment. The design of combinational logic circuits is arguably the most important topic in logic design and a thorough understanding of the principles of combinational design is an essential prerequisite to the design of more complex sequential systems, e.g. counters, registers and memories, and categorised as: asynchronous or synchronous. Programmable implementations may also be investigated.

Programmes

`COMP-0556`	BSc (Hons) in Applied Computing (`WD_KCOMP_B`)	2 / 4 / E

Indicative Content

- Number Systems.
- Boolean Algebra, Operators and Codes.
- Applications of combinational & sequential logic.
- Combinational and sequential implementations in digital circuits.
- Logic families.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain the fundamental concepts of combinational and sequential logic design.
2. Construct, simplify and convert Boolean expressions.
3. Define, analyse and design combinational and sequential logic circuits using SSI/MSI components.
4. Explain the differences between asynchronous and synchronous sequential circuits and implement counter and register applications.
5. Interpret data sheets and display a good understanding of the main characteristics of the different logic families.
6. Demonstrate the ability to build, simulate and test digital circuits.
7. Demonstrate a thorough knowledge of number systems, binary operations and codes.

Learning and Teaching Methods

- Lectures.
- Practicals / Project.
- Integrated demonstrations and examples.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,4,7
Continuous Assessment	40%	
Practical	40%	3,5,6

Assessment Criteria

<40%: Unable to interpret and describe key concepts of the specific knowledge domain(s).
40%-49%: Be able to interpret and describe key concepts of the specific knowledge domain(s).
50%-59%: Ability to discuss key concepts of the specific knowledge domain and ability to discover and integrate related knowledge in other knowledge domains.
60%-69%: Be able to solve problems within the specific knowledge domain(s) by experimenting with the appropriate skills and tools.
70%-100%: All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- "Orcad PCB Design Solutions." www.orcad.com
- "Programmable Logic. ." www.xilinx.com
- Floyd, T. . _Digital Fundamentals_ . 10th Ed. . NY: Prentice Hall, 2008.
- Lewin, D. and D. Protheroe. _Design of Logic Systems_. NY: Chapman & Hall, 1992.
- Wakerly, J. . _Digital Design, Principles and Practices_. 3rd Ed. . NY: Prentice Hall, 2005.

Supplementary Material

- Ashenden, P. _The Students Guide to VHDL._ . 2nd ed.. San Francisco: Morgan Kauffman, 2008.
- Katz, H. and G. Borriello. _Contemporary Logic Design_. 2nd ed. . New Jersey: Prentice Hall, 2005.

Requested Resources

- Lecture Room: Loose Seated
- ENGINEERING LAB: Electronics